

Unique decomposition Theorem.

For any $B \in GL_n(\mathbb{C})$, there exists a unique lower-triangular M with positive diagonal s.t. MB is unitary.

Proof. Since B is invertible, the row vectors of B , $\{\beta_1, \dots, \beta_n\}$ is basis for V .

Apply Gram-Schmidt process, that is, for each $1 \leq k \leq n$, put

$$d_k = \beta_k - \sum_{j < k} \frac{\langle \beta_k, d_j \rangle}{\|d_j\|^2} d_j$$

Then, $\text{span}\{d_1, \dots, d_r\} = \text{span}\{\beta_1, \dots, \beta_r\}$ by induction, and $\{d_1, \dots, d_n\}$ is orthonormal.

So, for each $1 \leq k \leq n$, $d_k = \beta_k - \sum_{j < k} C_{kj} \beta_j$ for some unique scalars $C_{kj} \in \mathbb{C}$.

Put $U = \begin{bmatrix} d_1 & \dots & d_n \\ \|d_1\| & & \|d_n\| \end{bmatrix}$, then clearly $U^* U = I$, because $\{d_1, \dots, d_n\}$ is orthonormal.

And $M_{kj} = \begin{cases} -\frac{C_{kj}}{\|d_k\|} & \text{if } j < k \\ \frac{1}{\|d_k\|} & \text{if } j = k \\ 0 & \text{otherwise.} \end{cases}$ Then, for each $1 \leq k \leq n$,

$$\|d_k\| \cdot \sum_{j=1}^n M_{kj} \beta_j = \beta_k - \sum_{j < k} C_{kj} \beta_j = d_k$$

So $MB = U$. To show the uniqueness, denote

$$T^+(n) = \{M \in M_{n \times n}(\mathbb{C}) \mid M \text{ is lower triangular, positive diagonals}\}.$$

Let $M_1, M_2 \in T^+(n)$ satisfies $M_1 B, M_2 B$ are unitary. Since unitary matrices forms a group

$$(M_1 B) \cdot (M_2 B)^{-1} = M_1 M_2^{-1} \text{ is also unitary.}$$

And, M_2^{-1} is lower-triangular with positive diagonal(s), so is $M_1 M_2^{-1}$. Now,

$$(M_1 M_2^{-1})^{-1} = (M_1 M_2^{-1})^*, \text{ so } M_1 M_2^{-1} \text{ is diagonal.}$$

So, $M_1 M_2^{-1} = I$, or $M_1 = M_2$. $\square$

Corollary. For each $B \in GL(n)$, there exists unique matrices K, A, N s.t. $B = KAN$, where K is unitary, A is diagonal with positive values, N is upper-triangular with diagonals 1

+Note. Let $T_n(\mathbb{C})$ be the set of $n \times n$ upper-triangular matrices with all diagonals are 1.

Then, $(T_n(\mathbb{C}), \cdot)$ is group, because: for any $A \in T_n(\mathbb{C})$, $A = I + N$,

where N is strictly upper triangular with $N^n = 0$, nilpotent. So,

$$A^{-1} = I - N + N^2 - N^3 + \dots + (-N)^{n-1},$$

and $A \in T_n(\mathbb{C})$ is clear.